

Two Analysts, One Planned Test

AP Statistics · Unit 3 · Topic 3.12 · B: 2026-12-15 · E: 2027-01-27 · TEACHER KEY

CED TETHER

- **Enduring Understanding VAR-6** — The normal distribution may be used to model variation.
- **Skill 1.F** — Identify null and alternative hypotheses.
- **Skill 1.E** — Identify an appropriate inference method for significance tests.
- **Skill 4.C** — Verify that inference procedures apply in a given situation.
- **LO VAR-6.H** — Identify the null and alternative hypotheses for a difference of two population proportions. [Skill 1.F]
- **EK VAR-6.H.1** — For a two-sample test for a difference of two proportions, the null hypothesis specifies a value of 0 for the difference in population proportions, indicating no difference or effect.
- **EK VAR-6.H.2** — The null hypothesis for a difference in proportions is: $H_0: p_1 = p_2$, or $H_0: p_1 - p_2 = 0$.
- **EK VAR-6.H.3** — A one-sided alternative hypothesis for a difference in proportions is $H_a: p_1 < p_2$ or $H_a: p_1 > p_2$. A two-sided alternative hypothesis is $H_a: p_1 \neq p_2$.
- **LO VAR-6.I** — Identify an appropriate testing method for the difference of two population proportions. [Skill 1.E]
- **EK VAR-6.I.1** — For a single categorical variable, the appropriate testing method for the difference of two population proportions is a two-sample z-test for a difference between two population proportions.
- **LO VAR-6.J** — Verify the conditions for making statistical inferences when testing a difference of two population proportions. [Skill 4.C]
- **EK VAR-6.J.1** — To make statistical inferences when testing a difference between population proportions, check for independence and approximately normal sampling distribution:
- Independence: data collected using two independent, random samples or a randomized experiment; when sampling without replacement, check $n_1 \leq 10\%N_1$ and $n_2 \leq 10\%N_2$.
- Shape (approximate normality): compute the combined (pooled) proportion $\hat{p}_c = (n_1\hat{p}_1 + n_2\hat{p}_2) / (n_1 + n_2)$; assuming H_0 is true, check that $n_1\hat{p}_c$, $n_1(1 - \hat{p}_c)$, $n_2\hat{p}_c$, and $n_2(1 - \hat{p}_c)$ are all ≥ 5 (or 10, per the predetermined value used).

Essential question: How do the original research question and study design determine a defensible two-proportion test setup?

DOK-3 skill: 4.C · **FRQ pattern:** predata-question-vs-postdata-tail-switch

DOK rationale: Part (c) coordinates the original question, timing of the tail choice, pooled conditions, and randomization scope to adjudicate two test setups and trace how the rejected setup distorts the evidence.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK: 1**
→ 3 **Minutes: 45**

Teacher does

- Board slide up at the bell. Ask which setup matches the team's question without confirming the tail.
- Begin the 6.10 video follow-along at minute 5.
- Return to the board slide at minute 33. Circulate and make students separate the question chosen before data from the sign observed afterward.

- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose Nia's or Omar's setup and cite wording or design evidence.
- Complete the follow-along about hypotheses, procedure selection, pooling, and conditions.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- What result would matter if the meter actually lowered completion?
- When was the direction of Omar's alternative chosen?
- Why are pooled expected counts appropriate under equality?
- What do random selection and random assignment each allow the team to claim?

Adult role

Solo teacher. Circulate 5 pairs. Require a setup, pooled-count check, and consequence of changing tails after seeing data.

[WARN] WATCH FOR

- Writing hypotheses with sample proportions instead of population parameters.
- Choosing a greater-than alternative only because the observed meter proportion is larger.
- Checking observed rather than pooled expected counts for the test.

SCORING PART (c) — E / P / I

Expected elements

- **selects-predata-two-sided-setup** (required) — Selects Nia and states a two-sided alternative
- **uses-question-and-timing** (required) — Uses the pre-data word change rather than the observed sign
- **cites-valid-conditions** (required) — Cites randomization, 10 percent, or pooled-count evidence
- **states-reader-consequence** (required) — Explains that a post-data tail switch hides decreases and exaggerates evidence
- **bounds-scope** (required) — Separates generalization from selection and causation from assignment

E	Chooses the two-sided setup; justifies its timing, conditions, consequence, and scope.
P	Chooses the two-sided setup but omits a condition, consequence, or scope statement.
I	Chooses a tail from the observed sign, uses sample statistics in hypotheses, or gives no design defense.

[STAR] TODAY'S PROBLEM — annotated

Suppose an online retailer has 50,000 active accounts. Before collecting data, its team asks, “Does a progress meter change the proportion of active accounts completing checkout?” It randomly selects 360 accounts and assigns 180 to each screen. The table shows the outcomes. Let p_M and p_S be the corresponding population completion proportions.

Nia: “Use $H_0 : p_M - p_S = 0$ versus $H_a : p_M - p_S \neq 0$ because either direction is a change.”

Omar: “Because $\hat{p}_M$ is larger, use $H_a : p_M - p_S > 0$.”

Checkout outcomes

Screen	n	Complete	Other
Meter	180	126	54
Standard	180	108	72

FIRST TAKE (5 min)

Which analyst’s setup should the team use? Cite one phrase or timing detail from the study.

Key: Not graded. The positive observed difference is intentionally tempting; the research question must control the tail.

Rules callout “SET THE QUESTION, PARAMETERS, AND CONDITIONS (keep on the page)” is printed on the student sheet.

DOK 1 (a) Define p_M and p_S in context. Calculate $\hat{p}_M$, $\hat{p}_S$, and the observed difference $\hat{p}_M - \hat{p}_S$.

Key: Here p_M and p_S are the completion proportions for all active accounts under the meter and standard screens. Thus $\hat{p}_M = 126/180 = 0.70$, $\hat{p}_S = 108/180 = 0.60$, and the difference is 0.10.

DOK 2 (b) Name the test procedure. Calculate the pooled proportion and verify randomization, the 10 percent condition, and all four pooled expected counts.

Key: The appropriate procedure is a two-sample z -test for a difference of proportions. The 360 accounts were randomly selected and assigned; $360 \leq 0.10(50,000) = 5,000$. Under H_0 , $\hat{p}_c = 234/360 = 0.65$, so each group has pooled expected counts 117 and 63, all at least 10.

DOK 3 (c) Decide which setup is defensible and state the hypotheses. Use the word ‘change,’ the timing of the tail choice, condition evidence, and both kinds of randomization. Explain how the rejected setup would misrepresent the strength or scope of the evidence.

Key: Nia is defensible: $H_0 : p_M - p_S = 0$ versus $H_a : p_M - p_S \neq 0$. The pre-data word ‘change’ includes either direction; the later difference 0.10 cannot select the tail. Random selection and assignment, $360 \leq 5,000$, and pooled counts 117 and 63 validate the procedure. The right-tail switch omits decreases and exaggerates evidence for a planned benefit. Selection supports generalization to accounts represented by the frame; assignment supports causation about screen type; neither extends beyond that frame.

EXIT

One thing the video changed about my first take: _____